

Role of the OSI Model in Software-Defined Networking (SDN)

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Indroduction

- OSI Model:

7-layer network communication model

- SDN:

Separates Control Plane and Data Plane

- Purpose:

Easy network management and automation

Role of OSI Layers in SDN

- ▶ **OSI Layer**
- ▶ **Role in Satellite Communication**
- ▶ Application (Layer 7)
 - Web browsing, email, video streaming
- ▶ Presentation (Layer 6)
 - Data encryption and formatting
- ▶ Session (Layer 5)
 - Maintains communication sessions
- ▶ Transport (Layer 4)
 - Reliable data delivery using TCP/UDP
- ▶ Network (Layer 3)
 - IP addressing and routing through satellites
- ▶ Data Link (Layer 2)
 - Error detection and frame transmission
- ▶ Physical (Layer 1)
 - Radio waves transmit signals between Earth stations and satellites

What is Software-Defined Networking (SDN)?

- ▶ SDN is a programmable networking approach.
- ▶ Centralized controller manages network devices.
- ▶ Improves flexibility and automation.
- ▶ Common protocol: **OpenFlow**.

Advantages & Conclusion

- ▶ Advantages:
- ▶ • Centralized control
- ▶ • Easy management
- ▶ • Better security
- ▶ • Lower cost

Conclusion

- ▶ OSI Model provides a structured networking framework.
- ▶ SDN separates the control plane and data plane.
- ▶ OSI layers help understand SDN operations.
- ▶ Improves network management and troubleshooting.
- ▶ Supports scalability, flexibility, and security.
- ▶ Forms the foundation for modern intelligent networks.